

Brendan Premo

Troy, NY | (518) 588-0855 | bdanp1998@gmail.com
linkedin.com/in/premobrendan | github.com/premobrendan/portfolio

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, Java, Excel, Jupyter, $\text{\LaTeX}$
- **Libraries & Tools:** PyTorch, Scikit-Learn, Pandas, GitHub, AutoGluon, Power BI (Sigma Computing)
- **Machine Learning:** Convolutional Neural Networks (CNN), Transfer Learning, Gradient Descent, Random Forest, Feature Engineering, Alternating Least Squares (ALS), Regression (Multivariate, Logistic, Time Series)
- **Data Analysis:** EDA, ETL, Statistics (Bayesian, Inferential and Nonparametric), KPIs, Trend Analysis, Variance Analysis, Validation
- **Mathematics:** Topological Data Analysis (TDA), Markov Chains, Monte Carlo Simulations, Principal Component Analysis (PCA), k-Rank SVD
- **Healthcare-Specific Analysis:** Claims, Readmissions, Length of Stay, Utilization, Costs, Insurance Plans (Medicaid, Medicare, Private), EMR/EHR, Medical Loss Ratio (MLR)

PROFESSIONAL EXPERIENCE

Soria Research

Data Analyst

June 2025 – April 2026

Remote

- Created the visual architecture of our products in Sigma, personally designed or refined over 75% of all client-facing analytical materials, specifically within the Healthcare sector.
- Orchestrated and developed end-to-end data pipelines, managing data acquisition, cleaning, and complex analysis for future use in the product, spanning EMR, EHR, and public health data, with emphasis on MLR, enrollment, and year-over-year deltas for companies and providers in the sector.
- Onboarded and trained new analysts on internal systems and data best practices.

EDUCATION

University at Albany

Albany, NY

Master of Science in Data Science

Fall 2023 – Spring 2025

Master of Arts in Mathematics

Fall 2022 – Fall 2024

Rensselaer Polytechnic Institute (RPI)

Troy, NY

Bachelor of Science in Applied Mathematics

Fall 2016 – Fall 2020

KEY TECHNICAL PROJECTS

Hospital Cost Report (Soria Analytics Dashboard, Sigma)

Winter 2025 – Spring 2026

- Created an interactive dashboard for the annual CMS Hospital Provider Cost Report dataset, breaking down national and provider-specific metrics (Revenue, Costs, Margins, Occupancy and Discharges).
- Set up a pipeline including a web crawler to automatically detect RSS feed updates for data, a scraper for automatic download, schema detection protocols to find deviations in formatting, and a system to incorporate the newly uploaded data directly into the dashboard.

Star Ratings (Soria Analytics Script, Python)

Fall 2025 – Winter 2025

- Developed a script that takes the raw Star Ratings tables from CMS and reconciles star ratings by plan.
- Analyzed plans individually, showing star rating changes from previous years as well as the metric shifts that caused the changes.
- Analyzed companies in aggregate by tracking changes in their plans' star ratings over time among NYSE-listed parent companies; Star Ratings are a well-known driver of trading interest, with share-price reactions often on the order of 5–10% depending on enrollment-weighted average star rating.

CDC Weekly Hospital Inpatient/ICU Occupancy (Soria Analytics Dashboard, Sigma)

Fall 2025

- Designed an in-depth dashboard showing Influenza-like Illness statistics over time, highlighting trends from season to season and allowing for aggregation based on varying timeframes.
- Includes optional breakdowns of trends based on age group to show differences in this demographic over time as well as in different regions (Census and HHS).
- Provides various visualizations of the data to suit individual preferences.
- Maintains a full end-to-end data pipeline so the dashboard updates each week when new data is released.

Biomechanics Topological Data Analysis (Data Science Thesis)

Spring 2025

- Analyzed movement patterns in sports video data to distinguish between optimal and suboptimal form.
- Transformed dynamic video into grayscale pixel data to perform Persistent Homology using Python, successfully mapping movement topology.
- Delivered a professional technical report and presentation summarizing complex topological findings for stakeholders.

Riemann Spectrum & Analytic Continuation (Mathematics Thesis)

Spring 2024

- Conducted extensive research on the history and formulations of the Riemann Hypothesis, focusing on analytic continuation.
- Developed and delivered a four-part lecture series to graduate students and professors, synthesizing complex theoretical mathematics into accessible presentations.

Machine Learning Projects: Predictive Modeling & Systems

Spring 2025

- **Housing Price Prediction:** Engineered a predictive model achieving a normalized Root Mean Square Error (RMSE) of 0.07 by utilizing AutoGluon's Tabular Predictor; performed extensive data cleaning and feature engineering.
- **Recommendation System:** Developed a movie/music recommendation engine from scratch, implementing Alternating Least Squares (ALS), Stochastic Gradient Descent (SGD), and Projected Gradient algorithms without relying on pre-built libraries.
- **Image Classification (CNN):** Achieved 85% classification accuracy on the CIFAR-10 dataset by implementing Convolutional Neural Networks; utilized transfer learning with AlexNet to demonstrate incremental network improvements.
- **Time Series Regression:** Conducted analysis on environmental factors, identifying a correlation where higher temperatures and lower precipitation serve as markers for higher Air Quality Index (AQI) levels.